Capstone Lab Task 2: WiFi

The objective of the Second task in the Capstone lab is to connect to a WiFi Network as a station. Therefore, we need to know the SSID (network name) and the network password. You can create an Accesspoint using your phone or laptop hotspot. Note that the ESP32 cannot connect to the university WiFi network, as it uses WiFi Enterprise, which requires a username and password.

Before implementing this task, it is essential to become familiar with the basic operation of the ESP32 WiFi Module. Generally, this module works in 2 major modes: Station (STA) and SoftAP. We are going to initialise the module in Station mode. Additionally, ESP32 supports IEEE 802.11 B/G/N at 2.4 GHz (not 5 GHz) and the TCP protocol. Thus, the ESP32 will obtain an IP address after the connection.

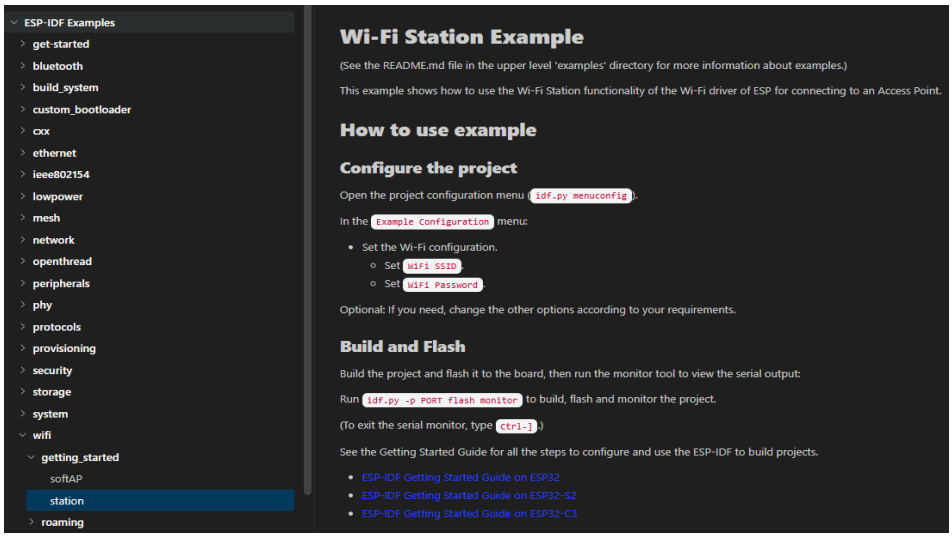
We use events for WiFi connections on the ESP32 because WiFi operations are asynchronous. When the ESP32 connects to a WiFi network, many steps happen in the background. These steps do not complete instantly and do not occur in a fixed time. Instead of blocking the program and waiting, the ESP-IDF WiFi stack generates events to notify the application of essential events. It is recommended that the following reference be reviewed before proceeding with this document.

https://docs.espressif.com/projects/esp-idf/en/latest/esp32/api-guides/wifi-driver/overview.html

In ESP-IDF, the WiFi transmission power can be set to optimise overall system power consumption. The link below provides useful information on configuring ESP wireless transmission power.

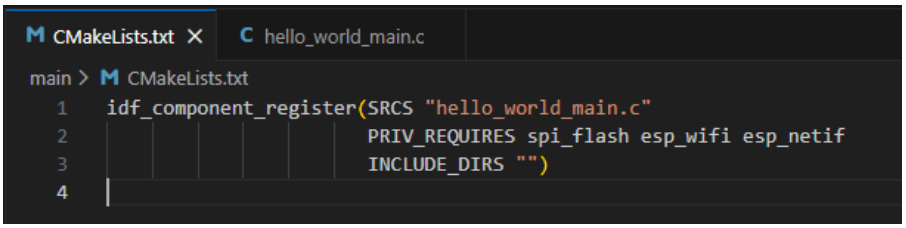
https://docs.espressif.com/projects/esp-techpedia/en/latest/esp-friends/advanced-development/performance/modify-tx-power.html

Similar to the previous task, a new project must be created. This project can be generated using either the Hello World template or the WiFi example projects in the path below in templates: WiFi →getting_started (station provided in the ESP-IDF).



The Hello World template is a more suitable starting point for this task, as it does not include additional Wi-Fi-related variables or functions that are not required for the implementation.

You will need the WiFi and network libraries in the project. Therefore, open the CMakelists.txt file under the main folder (not the one for the whole project) and add esp_wifi and esp_netif after PRIV_REQUIRES.



Additionally, esp_wifi.h and esp_netif.h header files should be added to the main file.

Finally, please note that the device uses 2 MB of flash memory by default. However, our ESP32 board has 4 MB of flash memory. You can change the device hardware configuration according to the “Getting Started” document.

Task 2. a:

Write a program to connect to a WiFi network. It should try to reconnect if the connection fails.

Test the connection and print the device's IP address and the security protocol used (e.g. WPA2) in the device terminal.

Task 2.b:

Change the maximum WiFi tx power to 10 dBm and test the connection again.